

The grapefruit: an old wine in a new glass? Metabolic and cardiovascular perspectives



Summary Grapefruit is a popular, tasty and nutritive fruit enjoyed globally. Biomedical evidence in the last 10 years has, however, shown that consumption of grapefruit or its juice is associated with drug interactions, which, in some cases, have been fatal. Grapefruit-induced drug interactions are unique in that the cytochrome P450 enzyme CYP3A4, which metabolises over 60% of commonly prescribed drugs as well as other drug transporter proteins such as P-glycoprotein and organic cation transporter proteins, which are all expressed in the intestines, are involved. However, the extent to which grapefruit–drug interactions impact on clinical settings has not been fully determined, probably because many cases are not reported. It has recently emerged that grapefruit, by virtue of its rich flavonoid content, is beneficial in the management of degenerative diseases such as diabetes and cardiovascular disorders. This potentially explosive subject is reviewed here.